QP CODE : 24900178

Discipline Specific Core Course : MG1DSCMBG101- WONDERS OF THE MICROBIAL WORLD

ANSWER KEY

PART -A

1. *Saccharomyces cerevisiae* (1mark)
2. Rhizobium (1mark)
3. Louis Pasteur (1mark)
4. 1.5 (1mark)
5. Hanging drop technique (1mark)
6. Protection and adherence (1mark)
7. Half spiral (1mark)
8. N-acetyl glucoseamine and N-acetyl muramic acid (1mark)
9. Bacillus stearothermophilus (1mark)
10. Pressure gauge (1mark)

Part B

11. Streptococci (1mark)
12. Protein synthesis (1mark)
13. Robert Koch (1mark)
14. Resolution (1mark)
15. Thick (1mark)

PART- C

16. False (1mark)
17. True (1mark)
18. False (1mark)
19. False (1mark)
20. True (1mark)

PART – D

21. Alcohol/ethanol (1mark)
22. Yoghurt (1mark)
23. Tetracyclin, Erythromycin, Penicillin etc (any two- 0.5 mark each)
24. Process of enlarging the appearance of an object when viewed through a microscope. (1mark)
25. Oil has a similar refractive index to glass - this reduces light refraction and allows more light to enter the lens - improves image clarity (1mark)
26. Any one - Mercury Vapour Lamps/Mercury arc Lamp/Xenon Lamps/LEDs etc (1mark)
27. Oxidation of cell components/ denaturation/ coagulation of protein (1mark)
28. Spherical cocci arranged in chains. (1mark)
29. Mycoplasma (1mark)
30. Protect from harsh environment, solute and protein transloction (1mark)

31. Proteins in the outer membrane of gram negative bacteria, control the uptake of small molecules. (1mark)
32. Made of asbestos used to filter microorganisms. (1mark)

PART – E

33. Role of *Thiobacillus ferroxidans, Pseudomonas putida* and others (3marks)

34. Arguments by supporters of spontaneous generation theory, Scientific proof for theory of biogenesis by Redi, Tyndall, Schulze and Schwann, Louis Pasteur (in brief) (3marks)

35. Parts of bright field microscope and functions of each part. (3marks)

36. Treat fungal infections, Azoles, Polyenes, Imidazole, Flucytosine etc , their uses and mode of action (any three antifungal agents ; 1 mark each)

37. Assay to test susceptibility of bacteria to an antibiotics. Filter paper disk with antibiotic placed on agar plate with bacteria, zone of inhibition measured. (3marks)

38. Potato dextrose agar or anyother fungal medium - inoculation-incubation. (3marks)

PART -F

39. Intracytoplasmic features -cytoplasm and cytoplasmic membrane **(2marks**), mesosomes **(1mark)** nucleoid **(1mark**) , plasmid **(1mark**), ribosomes**(1mark**), inclusions **(2marks**) etc

40. Definition **(1mark**) Principle : Acid-fast bacteria possess a thick, waxy layer composed of mycolic acids which makes them resistant to decolorization by acids and alcohol - Acid fast positive is red in colour, non-acid fast is blue in colour (**3 marks**) - Step-by-step procedure of acid fast staining (**4marks**)